Madrid Transport Analysis 2022

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1 Introduction and Sources

This project analyzes how meteorological conditions and the labor calendar affect urban mobility in Madrid during 2022, specifically the use of the BiciMad system and vehicular traffic.

The objective is to generate insights for sustainable mobility policies by analyzing transport patterns based on external factors such as climate and type of day.

Four official data sources from 2022 were used:

BiciMad: Trip records of the public bicycle system (EMT Madrid). More than 4 million journeys with information on dates, times, duration, and stations.
Fuente

Traffic: Hourly measurements of vehicular intensity from 60 permanent gauging stations (Madrid City Council).
Fuente

Meteorology: Daily climatic data including temperature, precipitation, wind, and pressure (Meteostat API).
Fuente

Labor Calendar: Official classification of working days, holidays, and weekends (Madrid City Council).
Fuente

The methodology includes three phases: individual exploratory analysis of each dataset, integration into a unified dataset for a preliminary joint analysis in which we check the coherence between datasets, and modeling in Power BI for interactive visualization.

2 Exploratory Data Analysis (EDA)

Exploratory data analysis constitutes the fundamental phase for understanding the structure, quality, and characteristics of each dataset before its integration. The main objectives of this stage are: identifying and correcting data quality issues (missing values, duplicates, anomalies), standardizing formats and nomenclature to facilitate subsequent integration, and extracting preliminary insights on patterns and distributions in each data source.

Each dataset presented specific challenges that required adapted solutions. Detailed analyses, including exhaustive technical explorations and additional visualizations, are documented in the individual notebooks for each EDA. This report presents only the most relevant aspects and key decisions made during the process.

2.1 BiciMadrid EDA

The BiciMadrid dataset consisted of 12 monthly CSV files with over 8.3 million initial records, representing all trips of the public bicycle system during 2022.

Identified Problems and Solutions:

- *Structural inconsistency*: Monthly files showed differences in the number of columns due to the irregular presence of the 'idTrip' column. This column was removed as it did not provide analytical value, allowing for the uniform concatenation of all files.
- *Interspersed empty rows*: 4.1 million completely empty rows were detected, distributed randomly in the dataset, resulting from errors in the generation of the source files. All were automatically eliminated.
- *Inconsistent station nomenclature*: The same physical point appeared with multiple names (e.g., "96 - Cibeles" vs "Cibeles", "Plaza España A" vs "Plaza España B"), artificially creating duplicate stations. Nomenclature was unified by eliminating numerical prefixes and A/B suffixes, consolidating from 400+ names to 258 real stations.
- *Duplicate records*: 140,000 trips (3.4%) were completely duplicated. They were removed, keeping a single instance of each trip.
- *Duration anomalies*: 16,256 records were identified with impossible durations (negative values up to -8,677 minutes and trips up to 696 hours). These anomalies were corrected by substituting them with the mean of valid values.
- *Missing values*: Station names presented ¡0.5% missing values, which were categorized as "Unknown" to preserve trip information.

Final Dataset: 4.07 million valid trips with 5 essential variables: duration, origin/destination stations, date, and time.

2.1.1 BiciMadrid Insights

The analysis revealed typical urban mobility patterns: average trip duration of 12.1 minutes, concentration on 5-15 minute journeys, and balanced distribution of use among stations (the top 10% concentrates only 20% of trips). These patterns confirm that BiciMadrid mainly functions as a complement to public transport for short connections.

2.2 Traffic EDA

The traffic dataset integrates 12 Excel files with measurements from 60 permanent stations in hourly format.

Identified Problems and Solutions:

- *Non-analytical structure*: The data came in horizontal format (hourly blocks and directions). They presented a structure of 4 rows per day and station (2 directions $\times$ 2 hourly blocks) with columns HOR1-HOR12 representing 2-hour slots each. The FSEN field simultaneously encoded traffic direction and hourly block ("1-" = Direction 1 hours 1:00-12:00, "1=" = Direction 1 hours 13:00-24:00, etc.). **Solution:** Application of the melt operation to convert columns HOR1-HOR12 into rows, followed by complex temporal mapping where the numerical index of the slot was extracted and the final hour was calculated according to FSEN (for "-" use 1-12 directly, for "=" add 12 resulting in 13-24), normalizing hour 24 to 0 to maintain the standard 0-23 range.
- *Hourly encoding in two blocks*: The separation into two halves of the day complicated the analysis. It was normalized to hours 0–23 to facilitate cross-referencing with other datasets.
- *Incomplete coverage*: 11 consecutive days in December are missing. The limitation is documented, and the daily analysis is performed on 354 valid days.
- *Very high values*: Peaks near 9 000–10 000 veh/h appear in several slots. They are not interpreted as errors or 'codes,' but as possible sensor ceilings. They were kept as is to respect the original record; their presence is infrequent and does not alter the general conclusions.

Final Dataset: 467,280 restructured rows with 4 variables: day, hour, station, and flow (aggregated by directions).

2.2.1 Traffic Insights

Realistic hourly patterns with right queues (peaks on main roads). Values close to 10,000 veh/h are consistent with measurement limits; the vast majority of observations are well below, so the extremes do not distort the set.

2.3 Meteorological Data EDA

The meteorological dataset was obtained through the Meteostat API, providing daily data for Madrid throughout 2022 with standard climatic variables.

Identified Problems and Solutions:

- *Direct download from API*: The data were obtained automatically via the Meteostat API using the coordinates of Puerta del Sol (40.4169, -3.7033) as a reference. The structure came predefined and normalized from the source, avoiding typical manual file format issues.
- *Irrelevant variables*: The download originally included 10 variables (`tavg`, `tmin`, `tmax`, `prcp`, `snow`, `wdir`, `wspd`, `wpgt`, `pres`, `tsun`), many with extensive missing values. Only the 4 complete and relevant variables were kept: average temperature (`tavg`), minimum (`tmin`), maximum (`tmax`), and precipitation (`prcp`).
- *Perfect temporal coverage*: The dataset covered exactly the 365 days of 2022 without temporal gaps, unlike other project datasets that had missing days.
- *Absence of anomalies*: No duplicate, missing values, or anomalies were detected in the selected variables. Extreme values in precipitation (up to 26.6 mm) and temperature (-0.5°C minimum, 40.7°C maximum) are climatologically coherent for Madrid.
- *Categorization for analysis*: Categorical variables were created using quartiles for temperature (Cold/Cool/Mild/Hot) and special separation for precipitation (No rain/Light rain/Heavy rain) to facilitate subsequent analysis of mobility impact.

Final Dataset: 365 daily records with 7 variables: date, 4 numerical climatic, and 2 derived categorical.

2.3.1 Meteorological Insights

Madrid 2022 registered an annual average temperature of 16.50°C (range: 3.40°C to 33.70°C) with marked seasonality. Precipitation was scarce: average of 1.33 mm/day, and it was found that there were nearly 300 days of the year without rain (0 mm). The maximum precipitation was 26.60 mm. The data reflect the typical continental Mediterranean climate, providing sufficient variability to analyze impacts on urban mobility patterns.

2.4 Labor Calendar Data EDA

The labor calendar dataset was obtained from the Madrid City Council's Open Data Portal, containing the official classification of days for multiple years (2013-2024).

Identified Problems and Solutions:

- *Excessive temporal coverage*: The original dataset contained 4,747 records covering multiple years (2013-2024), when we only needed 2022. A specific filter by year was applied, resulting in exactly 365 days of 2022.
- *Irrelevant columns*: The file included columns `Tipo de Festivo` and `Festividad` with detailed information on celebrations that is not relevant for mobility analysis. They were removed, keeping only the essential variables.
- *Character encoding*: The first column presented encoding characters (`ï»¿Dia`). It was correctly renamed to `Dia` to prevent later errors.
- *Incomplete data in Type column*: After filtering to 2022, the `Tipo` column only contained `Festivo` values for official holidays, but missing values (NaN) for the rest of the days. Automatic mapping was implemented using `Dia_semana`: Monday-Friday → `laborable`, Saturdays → `sabado`, Sundays → `domingo`.

Final Dataset: 365 records with 3 variables: date, day of the week, and categorical day type (working day/Saturday/Sunday/holiday).

2.4.1 Labor Calendar Insights

The 2022 analysis reveals the distribution of the Madrid labor calendar: approximately 250 working days (non-holiday Monday-Friday), 52 Saturdays, 52 Sundays, and the remaining official holidays. This categorization provides the fundamental basis for analyzing how patterns of economic and social activity (working days vs. weekends vs. holidays) impact urban mobility behaviors for both BiciMad and vehicular traffic.

3 Integration and Joint Analysis

This section consolidates the four individual datasets into a unified analysis that allows for the examination of the relationships between urban mobility, meteorological conditions, and the labor calendar.

3.1 Data Aggregation

The integration process required transforming datasets with different temporal granularities to a common daily level:

Aggregation by day: BiciMad and traffic data, originally with hourly granularity, were aggregated by summing by day. BiciMad became the total number of daily trips, while traffic was transformed into total daily vehicular flow. Meteorological and labor calendar data were already at daily granularity.

Fusion using inner join: An inner join strategy by date was applied to keep only the days present in all datasets. This approach guaranteed temporal integrity but limited the analysis to 354 days due to the 11 missing days in the traffic dataset (December 1-11, 2022).

Calendar Encoding: An ordinal variable `tipo_numerico` was created where working day=0, Saturday=1, Sunday/holiday=2, reflecting decreasing levels of work activity. The final distribution was: 242 working days (68%), 50 Saturdays (14%), and 62 Sundays/holidays (18%).

3.2 Correlation Analysis

The Pearson correlation matrix revealed significant patterns in Madrid urban mobility:

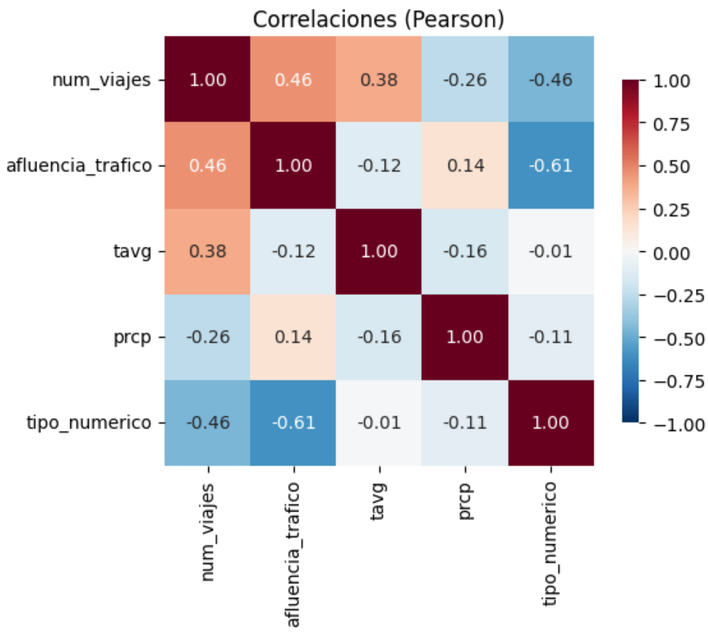

Figure 1: Correlation Matrix Heatmap

Impact of the labor calendar: The type of day showed strong negative correlations with both BiciMad (-0.46) and vehicular traffic (-0.61). Traffic shows greater sensitivity to the labor calendar, indicating that private vehicle trips are more linked to work activities than the use of public bicycles.

Differentiated meteorological effects: Temperature (tavg) positively affects BiciMad (+0.38) and negatively affects traffic (-0.12). Precipitation (prcp) shows the inverse pattern: negative for BiciMad (-0.26) and slightly positive for traffic (+0.14). This suggests that high temperatures favor urban cycling while rain discourages it, pushing towards vehicular transport.

Complementarity between modes: The positive correlation between BiciMad and traffic (+0.46) indicates that both respond to common patterns of daily mobility demand, functioning more as complements than as perfect substitutes in the urban transport ecosystem.

These results confirm that urban mobility is significantly influenced by both structural factors (labor calendar) and variables (meteorological conditions), with different sensitivities depending on the mode of transport.

4 Data Model and Metric Creation in Power BI

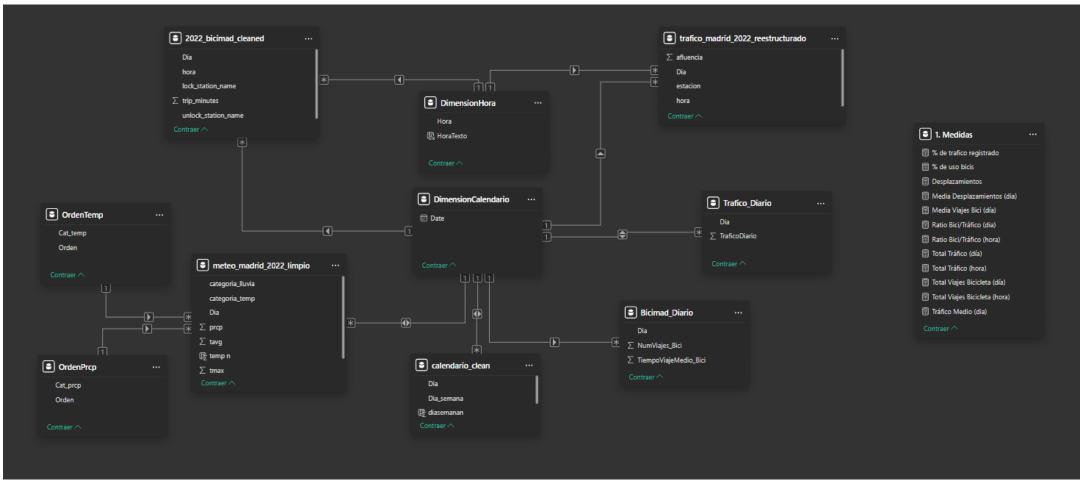

Figure 2: Relational Data Model and Metrics Table

General Architecture

The model implements a star schema with two granularity levels: daily and hourly. This structure allows for efficient aggregated analysis (daily trends) and simultaneous intra-day detail, unifying bicycles, motorized traffic, and meteorological-temporal context under common dimensions.

Daily granularity level

- **Central Tables**: `Bicimad_Diario` and `Trafico_Diario` (created using Group By in Power Query from the hourly data tables).
- **Dimension**: `DimensionCalendario` with key `Date`
- **Context**: integration with `meteo_madrid_2022_limpio` and `calendario_clean` by date
- **Purpose**: visuals of trends, monthly comparisons, and climate-mobility correlation analysis

Hourly granularity level

- **Central Tables**: `2022_bicimad_cleaned` (trip level) and `trafico_madrid_2022_reestructurado` (station-hour level)
- **Dimensions**: `DimensionHora` with key `Hora` (0-23) and `DimensionCalendario` with key `Date`
- **Purpose**: hourly usage patterns, peak hour identification, and daily seasonality analysis

Relationships and Filtering

Daily level tables connect only to `DimensionCalendario`, while hourly tables maintain a double connection: with `DimensionHora` for intra-day analysis and with `DimensionCalendario` for temporal coherence by date. This architecture ensures that date filters propagate consistently to both granularity levels, avoiding ambiguities in cross-analysis.

Key Transformations

- Daily aggregation in Power Query to optimize performance in trend visuals
- Creation of auxiliary sorting tables (`OrdenTemp`, `OrdenPrcp`) for meteorological categories
- Consistent typing of temporal keys (Date, integer) for stable relationships

4.0.1 Metric Creation

Metrics are organized into volume metrics (total traffic and trips by day/hour), central tendency (daily averages for temporal comparisons), and proportion (bike/traffic ratio and percentage of sustainable use). Metrics are centralized in the table "1. Measures".

5 Global Results and Insights

This analysis is derived from the dashboards generated in Power BI, where a deeper understanding can be achieved by interacting with the report's filters. In this section, we present the main conclusions derived from the in-depth analysis of the generated dataset.

5.1 General System Magnitudes

The joint analysis reveals a considerable annual volume of urban mobility: 719.5 million vehicular traffic records (daily average: 2.02 million) and 4.07 million bicycle trips (daily average: 11.16 thousand). These data reflect a bike/traffic ratio of 0.57%, indicating the complementary nature of bicycle transport compared to motorized transport.

5.2 Temporal Analysis

5.2.1 Monthly Variation

Vehicular Traffic

- Relative inter-monthly stability with a peak in March (66+ million)
- Moderate decreases in June-July and pronounced decrease in August (47 million)
- December presents abnormally low values due to the absence of 11 days of records

Bicycle Usage

- Marked seasonality: maximum in September (447 thousand trips) and minimum in August (270 thousand)
- The months May-September (except August) concentrate the highest bicycle use
- Significant reduction in winter months, consistent with climatic preferences

5.2.2 Intra-day Distribution

Differentiated patterns by transport mode

- **Common nocturnal trough**: minimum activity between 00:00-07:00 in both modes

- **Morning activation**: abrupt increase at 08:00 coinciding with the start of the work day
- **Vehicular Traffic**: sustained level from 08:00 until the end of the day
- **Bicycles**: three differentiated peaks (08:00, 14:00, 19:00) aligned with work hours (start, lunch break, end)

This distribution suggests that bicycle use functions as complementary transport oriented towards specific work trips, while vehicular traffic maintains constant demand throughout the day.

5.3 Impact of Day Type

5.3.1 Activity Differences

Mobility presents a clear difference according to the nature of the day:

- **Working days**: maximum activity (bikes: 12.3 thousand; vehicles: 2.2M daily)
- **Saturdays, Sundays, and holidays**: minimum activity (bikes: 6.8 thousand; vehicles: 1.4M)

The holiday/working day difference reaches 55% for bicycles and 63% for traffic, confirming the dependence of both modes on economic activity.

5.3.2 Hourly Reconfiguration on Non-Working Days

Weekends and holidays substantially modify intra-day patterns:

- Significant delay in the start of activity (from 08:00 to 10:00-12:00)
- Concentration in two time slots in both transport modes: 12:00-14:00 and 18:00-20:00
- Disappearance of the three differentiated peaks of work-related cycling use

5.4 Meteorological Influence

5.4.1 Temperature

Vehicular Traffic: minimum impact in mild, cool, and cold categories (around 2M average). The reduction on hot days (1.87M average) might not be due to causality, but to the coincidence with summer months when there is less activity, especially in August.

Bicycle Usage: clear preference for mild temperatures (average of 13.8 thousand trips), followed by hot ones. Notable reduction in cool and especially cold conditions, with an average of only 8.6 thousand trips per day, consistent with the May-September concentration observed monthly.

5.4.2 Precipitation

Marked inverse effect between modes, as already detected in the correlation analysis, where a negative correlation (-0.26) was observed between rain and bicycles, while positive (0.14) with traffic:

- Increase in vehicular traffic with greater precipitation
- Proportional decrease in bicycle usage

5.5 Interaction Insights

5.5.1 Prevalence of the Labor Factor

The scatter plots by day type and temperature reveal that **the labor nature of the day surpasses meteorological factors in impact**. Across the entire range of average temperatures, working days systematically maintain higher activity than Saturdays, Sundays, and holidays, confirming what was observed in the previous joint analysis, where the correlation observed with the `tipo_numerico` variable (-0.46 bikes, -0.61 traffic) had greater absolute values than those associated with temperature (0.38 bikes, -0.12 traffic).

5.5.2 Modal Complementarity

The positive correlation between bicycles and traffic (0.46) indicates a common response to urban mobility patterns. In every temporal analysis and with any filter applied, both graphs behave similarly, suggesting a systematic complementarity between transport modes and confirming the correlation observed in the previous correlation analysis. However, the different intra-day profiles reveal differentiated roles: traffic as the continuous base mode and the bicycle as a complement for specific trips.

6 Conclusions

This study has successfully integrated urban mobility, meteorology, and labor calendar data for Madrid in 2022, generating a comprehensive analysis of sustainable and motorized transport patterns in the city.

6.1 Methodological Achievements

The project has demonstrated the viability of fusing heterogeneous urban data sources through cleaning, normalization, and temporal aggregation techniques. The data architecture implemented in Power BI, with double granularity (daily and hourly), allows for both aggregated trend analysis and detailed intra-day pattern analysis.

6.2 Main Findings

6.2.1 Determining Factors of Mobility

The analysis reveals that **the labor nature of the day constitutes the most influential factor** in both transport modes, surpassing meteorological conditions in impact. Working days systematically record higher activity than weekends and holidays, regardless of climatic conditions.

6.2.2 Complementarity between Transport Modes

The positive correlation (0.46) between bicycle use and vehicular traffic, along with the similar behavior observed in all temporal analyses, confirms a **systematic complementarity** between both modes. However, they present differentiated roles: vehicular traffic maintains constant demand throughout the day, while bicycles show concentrated use during specific work hours.

6.2.3 Differentiated Meteorological Sensitivity

Bicycles exhibit greater sensitivity to meteorological conditions than vehicular traffic. Temperature favors cycling use (correlation 0.38) and precipitation penalizes it (-0.26), while vehicular traffic shows less climatic variability.

6.3 Implications for Mobility Policies

The results suggest opportunities for sustainable mobility policies:

- **Workplace Promotion**: given the strong link between cycling use and work activity, corporate sustainable mobility initiatives can have a high impact.
- **Resilient Infrastructure**: the meteorological sensitivity of cycling use justifies investments in protected infrastructure (covered lanes, sheltered parking).
- **Modal Complementarity**: the parallel behavior of both modes suggests that vehicular restriction policies can effectively boost cycling use if accompanied by adequate infrastructure.

6.4 Limitations and Future Work

The study presents limitations that open up lines for future research. The absence of 11 days of traffic data in December affects temporal representativeness. The loss of spatial granularity in the joint analysis limits insights into the territorial distribution of mobility.

Future research could incorporate data from other public transport modes (metro, bus) for a complete view of the mobility system, analyze the price-demand elasticity of cycling transport using fare data, and develop predictive demand models that integrate multiple contextual variables. Incorporating data from other cities would also be useful in obtaining a complete picture of how meteorological and calendar factors affect urban mobility in different geographical and socioeconomic contexts.